Sri Lanka Institute of Information Technology

PROJECT REGISTRATION FORM

(This form should be completed and uploaded to the Cloud space on or before XXXXXXXXX)

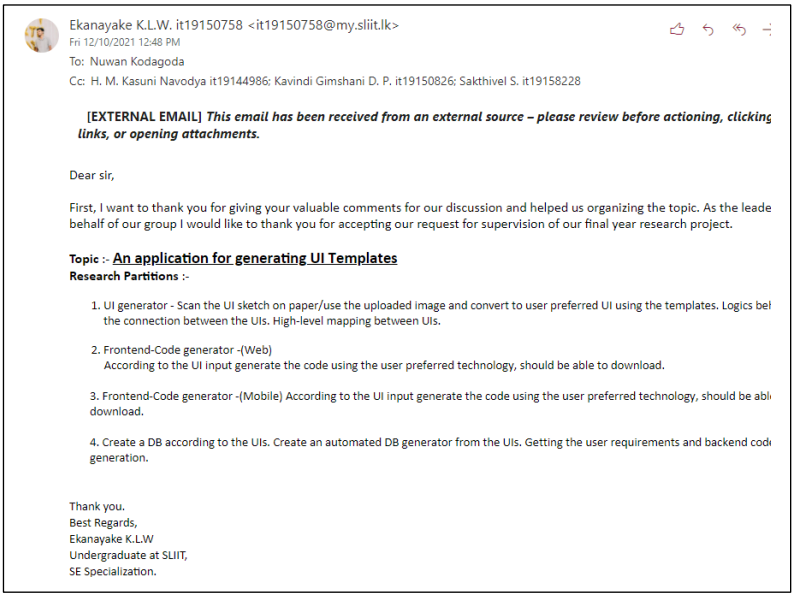
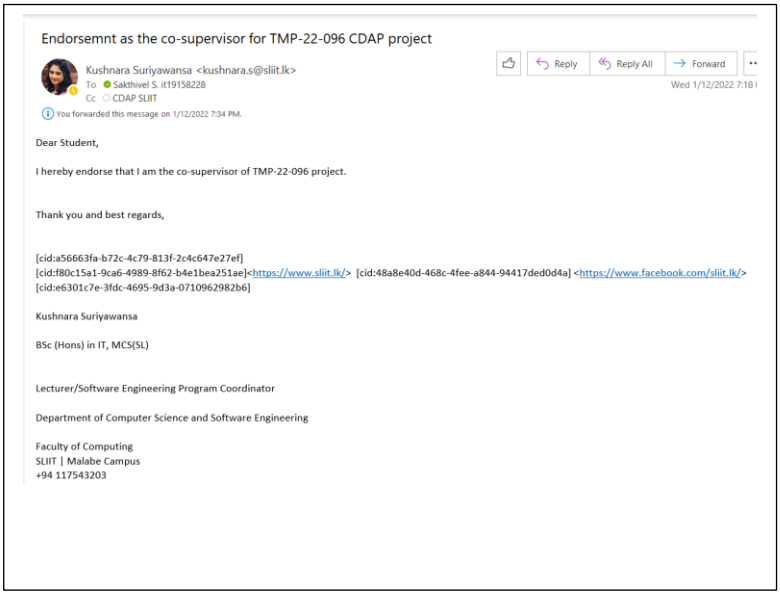
The purpose of this form is to allow final year students of the B.Sc. (Hon) degree program to enlist in the final year project group. Enlisting in a project entails specifying the project title and the details of four members in the group, the internal supervisor (compulsory), external supervisor (may be from the industry) and indicating a brief description of the project. The description of the project entered on this form will not be considered as the formal project proposal. It should however indicate the scope of the project and provide the main potential outcome.

PROJECT TITLE (As per the accepted topic assessment form)	**UI Builder: A tool with enhanced features to build, design and develop UI along with automatic code generation for software developers & managers.**
RESEARCH GROUP **(As per the Topic assessment Form)**	Software Systems & Technologies (SST)
PROJECT NUMBER	(will be assigned by the lecture in charge)

PROJECT GROUP MEMBER DETAILS: (Please start with group leader's details)

	STUDENT NAME	STUDENT NO.	CONTACT NO.	EMAIL ADDRESS
1	Ekanayake K.L.W (***GP LEADER)***	IT19150758	0766354287	it19150758@my.sliit.lk
2	Sakthivel.S	IT19158228	0762878750	it19158228@my.sliit.lk
3	H.M. Kasuni Navodya	IT19144986	0773459994	it19144986@my.sliit.lk
4	D.P Kavindi Gimshani	IT19150826	0757970341	it19150826@my.sliit.lk

SUPERVISOR, CO_ SUPERVISOR Details

SUPERVISOR Name	CO-SUPERVISOR Name
Dr. Nuwan Kodagoda	Ms. Kushnara Suriyawansa
Signature	**Signature**
Appendix 1 	**Appendix 2** 
10th of December 2021	12th of January 2022
Date	**Date**

EXTERNAL SUPERVISOR Details (if any, may be from the industry)

				Attach the email as Appendix 3
Name	Affiliation	Contact Address	Contact Numbers	Signature/Date

ACCEPTANCE BY CDAP MEMBER (This part will be filled by the RP team)

Name	Signature	Date

PROJECT DETAILS

Brief Description of your Research Problem: (extract from the topic assessment form)

The main problems identified for the research are listed in three different sections. The first section focusses mainly on the problems faced by the user, whereas the second section gives an overview about the problems faced by business. Finally, the last section includes the common problems in public/people with less/without knowledge in technology that needs to be addressed. The main problem that showed up when the topic was analyzed was the inability of UI designers and software developers to create UI according to their preferences. The user expected for a real-time display of their preferred UI with different templates which was lacking in most of the tools in the market.

While the user faced the above-mentioned problems, it was able to see the business did not take advantage from the existing tools in the market for their design and development. This issue was further investigated to find the underlying problems. The project managers expected to reduce the designing time from sketches to mockups. UI designers finds it difficult to get the real needs from users by showing parts of UI, which they need a complete UI design as alternatives.

Any tool that makes an impact on society has the maximum probability for being success and long-term existence on the market. We were able to collect some thoughts from non-technical people who was interested in a tool as mentioned above. We had to bridge the gap between the two extremes where the usability of the tool should remain the same for any user interacting with the tool. There was a strong need for non-technical or people with less technical knowledge expecting to develop small application for their business pr personal needs. Any user using our application shouldn't be shown the complexity behind the process he/she is trying to achieve. Usability and reliability of the application should be enhanced to get a higher conversion rate.

Main References:

- Queiros, Ricardo, "SeCOGen – A Service Code Generator", in 8th Symposium on Languages,Applications and Technologies (SLATE 2019) [Online]. Available: https://drops.dagstuhl.de/opus/volltexte/2019/10890/pdf/OASIcs-SLATE-2019-23.pdf

- A. A. Sutchenkov and A. I. Tikhonov, "Site Generator for Small Open and Private Online Courses," 2020 International Youth Conference on Radio Electronics, Electrical and Power Engineering (REEPE), 2020, pp. 1-5, doi: 10.1109/REEPE49198.2020.9059197.

- Watanabe, K., Imamura, M., Asami, K. and Amanuma, T., 2009. A Web Application Development Framework Using Code Generation from MVC-Based UI Model. *Distributed Computing, Artificial Intelligence, Bioinformatics, Soft Computing, and Ambient Assisted Living*, pp.404-411.

Description of the Solution: (extract from the topic assessment form)

A web application that can facilitate user including UI designers, frontend developers and database designers to automate their actions and save time. To be brief with the functionalities of the application proposed following features are expected to be packed inside the application.

1. Convert the sketch on paper to a standard UI considering the alignment of UI components. [with standard fonts and color]
2. Provide different templates of UI [both user-defined & pre-defined] templates. Allow user to pick templates while showing a real-time display for each selected template.
3. According to the template selected store the UI on the database for further processing.
4. According to the template selected generate the frontend code [Web/Mobile] against the technology preferred by the user.
5. According to the template selected generate XML Meta file which is used for the entire backend code generations (DB Creation, validation code generation, business logic embedded code)

Main expected outcomes of the project: (extract from the topic assessment form)

Main Objective:

- Making the software development process simple, efficient, and robust.
- A tool that can generate complete working source code with embedded business logic.

Sub Objective 1:

- Generate UI templates (User Preferred) for the UI sketches on the papers/uploaded images.

Sub Objective 2:

- Conversion of UI to XML Meta file for generation of backend code by considering the business logic, validation, and database connectivity through forward engineering systems (FES).

Sub Objective 3:

- Generate web frontend code using user preferred technologies according to the user requirements of the selected UI template.

Sub Objective 4:

- Generate the code for the mobile application UI by using the user preferred technology and platform.

WORKLOAD ALLOCATION (**extract from the topic assessment form after the correction suggested by the topic assessment panel.**)

(Please provide a brief description about the workload allocation)

MEMBER 1	**Ekanayake K.L.W - IT19150758**

Component - UI generator

Basic idea of this component -

Scan the UI sketch on paper/use the uploaded image and convert to user preferred UI using the templates. Logics behind the connection between the UIs. High-level mapping between UIs.

Expected tasks -

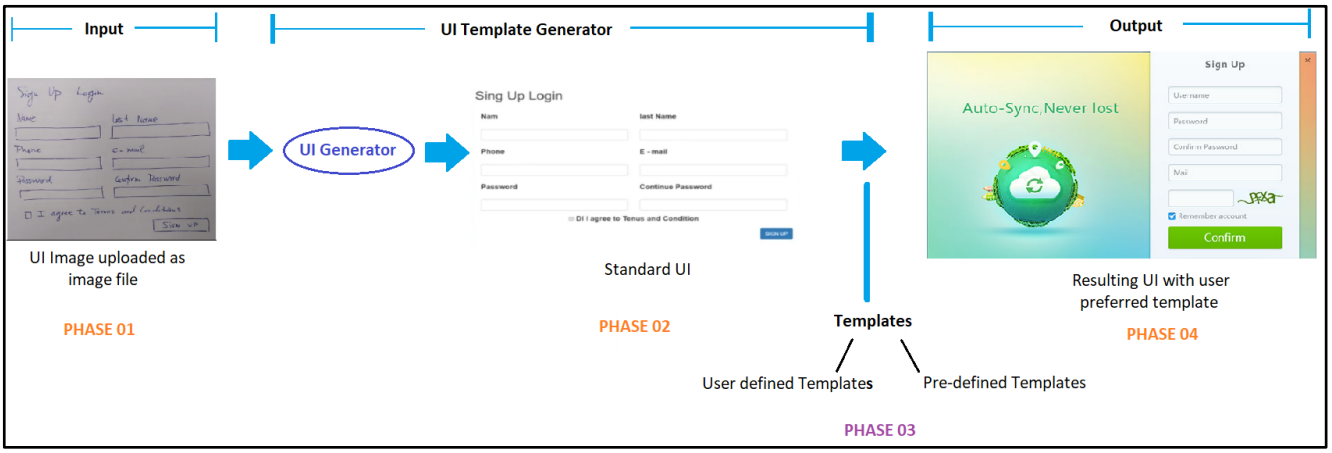
- Process the images captured/uploaded to the application.
- Identify the spacing, alignment, width, and height of components in the UI.
- Convert all written texts into a standard text and build up the UI.
- The application should be intelligent enough to find components that are not completed or missing in the UI.
- The standard UI should be first UI template for users.
- Provide different UI templates to user and allow him/her to select.
- Free-hand designing should also be improved where user can customize or a create a new template according to wish.

Major steps -

1. **Step 01**
   - The captured image is uploaded/directly captured through mobile and given to the application.
   - The image inputting process has some features.
     - Image contrast.
     - Image value [light/dark].
     - Image size.
     - Image noise reduction.
2. **Step 02**
   - The enhanced image from process 1 is converted to a standard UI.
   - The following will be considered:
     1. Identify the components in the sketch which are partially completed.
     2. The spacing, alignment of the components.
     3. Convert written words into standard fonts.
     4. Use standard colors on components.

3. **Step 03**
   - The standard UI resulted from process 2 is converted to UI needed by user.
   - There are two types of templates to be given for the user.
     - ✓ User-defined templates.
     - ✓ Pre-defined templates.
   - The user is expected to select a template or create a template.
   - According to the selected/created template the user will be able to see a real-time preview of the UI.
   - Finally, the template user selected/created will be mapped with the UI and given for code generation.

<u>Explanation on input output for the component</u> -



PHASE 01 – The UI image is uploaded to the application. This is a mockup where the UI components are sketched using pen and paper. The uploaded image is checked for its quality and optimized in a way to be processed.

PHASE 02 – The uploaded image is sent through the UI generator where the paper sketch is converted to a standard UI. The standard UI is the resulted-on PHASE 02. This UI lacks styles and colors. It will act as an intermediate skeleton of the UI where the templates can be added.

PHASE 03 – This phase involves in adding templates to the standard UI. The templates are of two types namely, user defined templates and pre-defined templates. The user is allowed to select or create new templates according to their preferences. The real-time preview on each selected template allows the user to decide which suits the UI best.

PHASE 04 – This will be final output from this research component. According to the user selected template the UI is designed and displayed to the user with styles and colors. The layout, spacing between UI elements, images and other visual elements are placed according to user preferences.

MEMBER 2	**Sakthivel.S - IT19158228**

Component – Backend code with embedded Business Logic & DB generator using forward engineering process

Basic concept of the component -

From the Component 1 the UI is created by the user. The UI designed by the system is converted to XML-Meta data. The XML-Meta data is used for generating code with business logic (BL), validation code generation and the DB generation.

Brief explanation about the process -

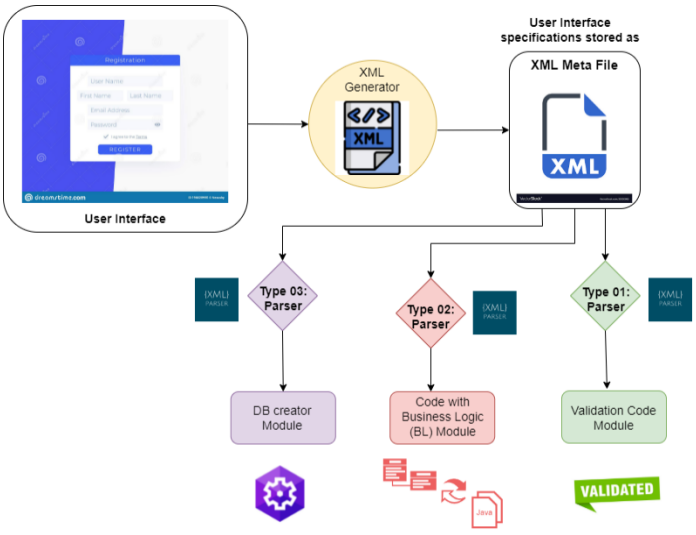
- As the initial step the UI resulted from the component 1 is converted to XML Meta file.
- Then from the XML Meta file obtained validation code, code with business logic and the DB is generated.
- Different parsers are used to achieve each of the above-mentioned code segments.
- 3 different types of XML parsers will be used along with 3 modules namely database creation module, Business Logic code generation module and validation code generation module respectively.
- Finally, the above 3 modules will be integrated and used as the final backend code for the application.

Objective from the component -

- Making the software development process simpler, efficient, and robust.

The reason for tackling forward engineering system (FES) -

- FES can support multiple languages (C, C++, C#, and java)
- Support of multiple DBMS (MySQL, SQL, MongoDB)

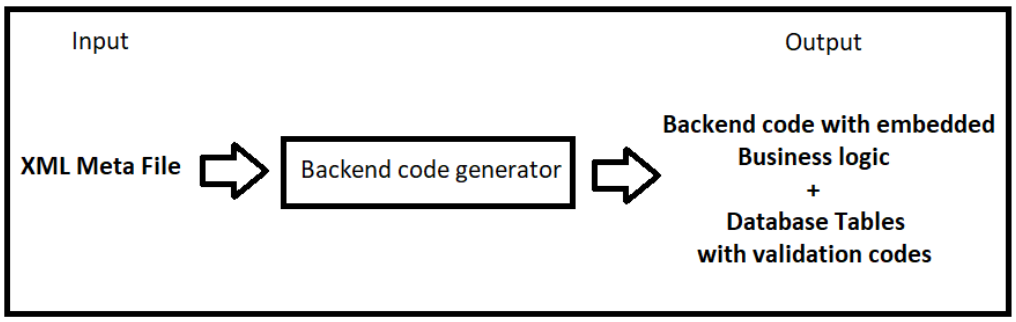


Background study on existing tools – Analysis -

Existing tools taken for the study: ORACLE, Microsoft VB (Main backend-oriented tools were selected)

- The main observation can be summarized as developers spending more time on code implementation for the same functionality without reusing the code.
- The need of backend developers with a database that is easily manageable with automated validation.

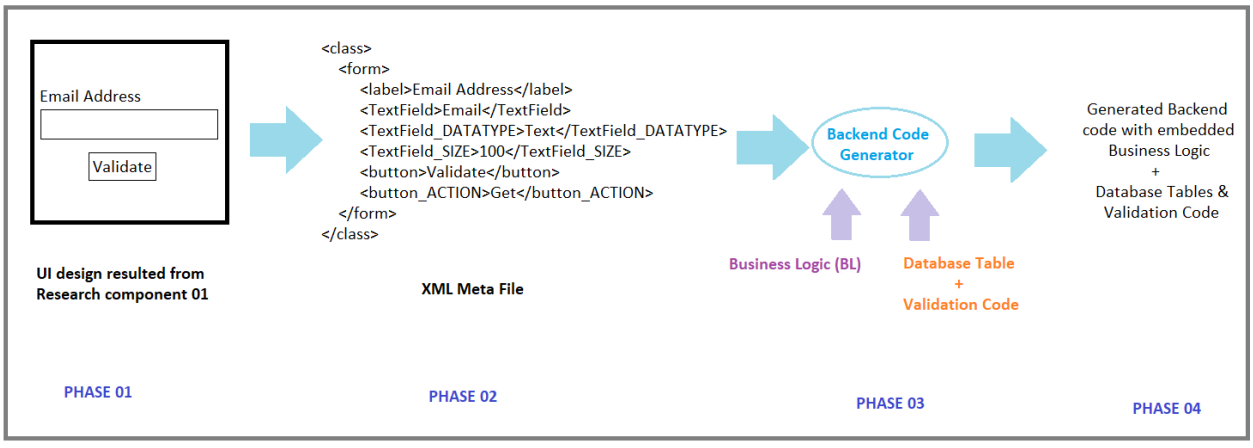
Explanation on input output for the component -



The UI specification is stored as XML Meta file and sent to the backend code generator as the input. The major attributes present in the UI are included on the XML meta file.

The output from the backend code generator will result in Business Logic embedded in the generated code along with database tables with validation code. Coding conventions, database validations and commenting of code will be added characteristics of the output.

Process – High level understanding -



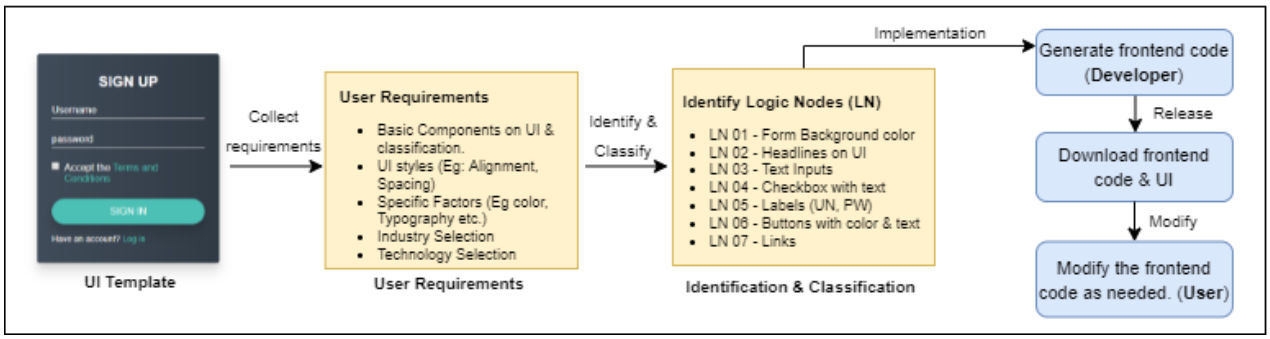
MEMBER 3	H.M. Kasuni Navodya – IT19144986

Component – Web Frontend-code generator using Logic Node Algorithms

This component intends to implement web frontend code to match the selected UI template according to the user's requirements.

The flow of the component –

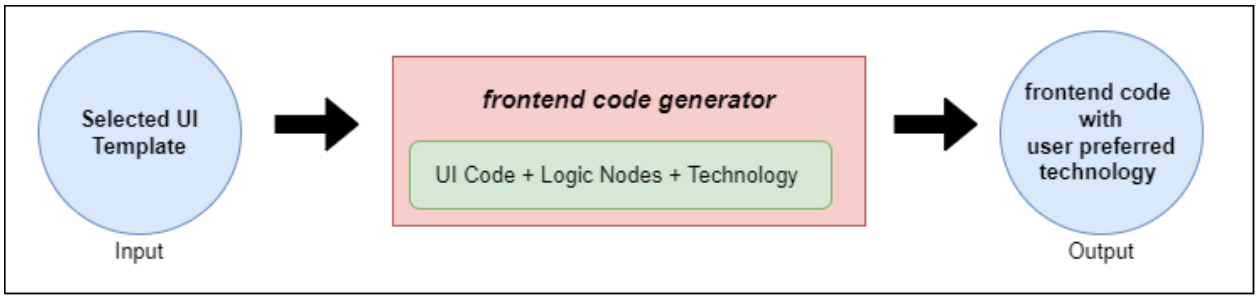
1. List the User requirements according to the UI template selected by the User.
2. Identify and classify the elements of selected UI template.
3. After classification, place the elements onto separate **Logic Nodes (LN)**.
4. According to the logic nodes, generate frontend code using user preferred technology.
    - NOTE: Consider the scope here and take steps to use 2 user preferred technologies.
5. Finally download the generated web frontend code with UI template.
    - NOTE: After downloading, the user can modify the web frontend code as required.



Novelty in the component –

1. Relationship between the UI fields ->
    - The identification of relationship between the fields in UI will be a newly added feature.
    - E.g.: Master-details, lookup relationships will be identified by the application intelligently before actual code is generated.
2. Technology ->
    - Expected to be selected by the user (Targeted to generate code for 2 technologies minimum)
3. Flexibility of the code ->
    - Changes are uncertain and the user may think code should be modified after code generation. Even after the code is downloaded it should be modified and UI should be flexible to respond to new changes.

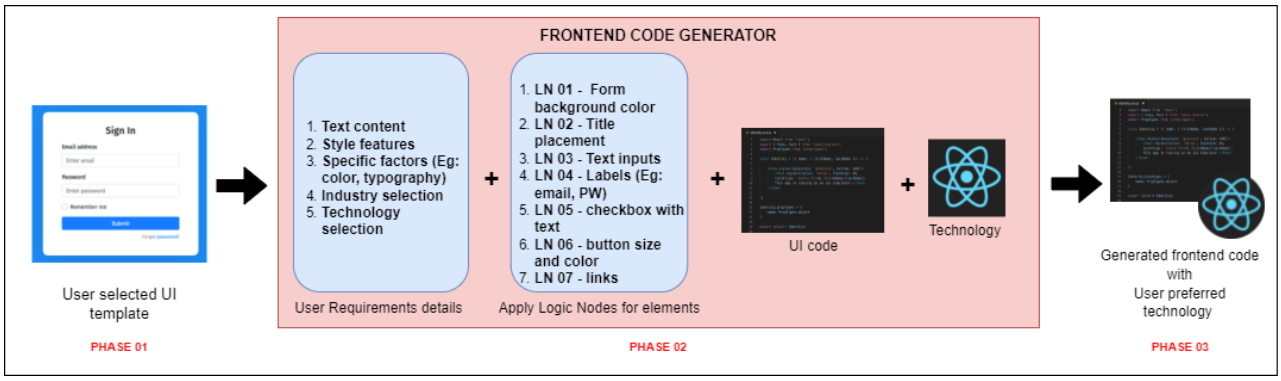
Explanation on input output for the component –



The result of component 1, the UI template is used as the **input**. The process of frontend code generator is to place the elements of the UI template onto separate logic nodes. Then the web frontend code is implemented using those logic nodes. Furthermore, the technology preferred by the user is also taken into consideration while implementing the front code here.

Finally, the frontend code generator releases the web frontend code with user preferred technology as a **result**.

Process – High level understanding -



Uses and Benefits –

1. It is important to increase the productivity of the development process.
2. Generate the frontend code from the description. This helps to understand the logic behind the code, even when the code is complex.
3. Generated code will follow a fixed coding standards according to organization/user preference. It is likely that many errors occur in the code when multiple developers work on the same code base. This can be prevented when front end code generators are used.

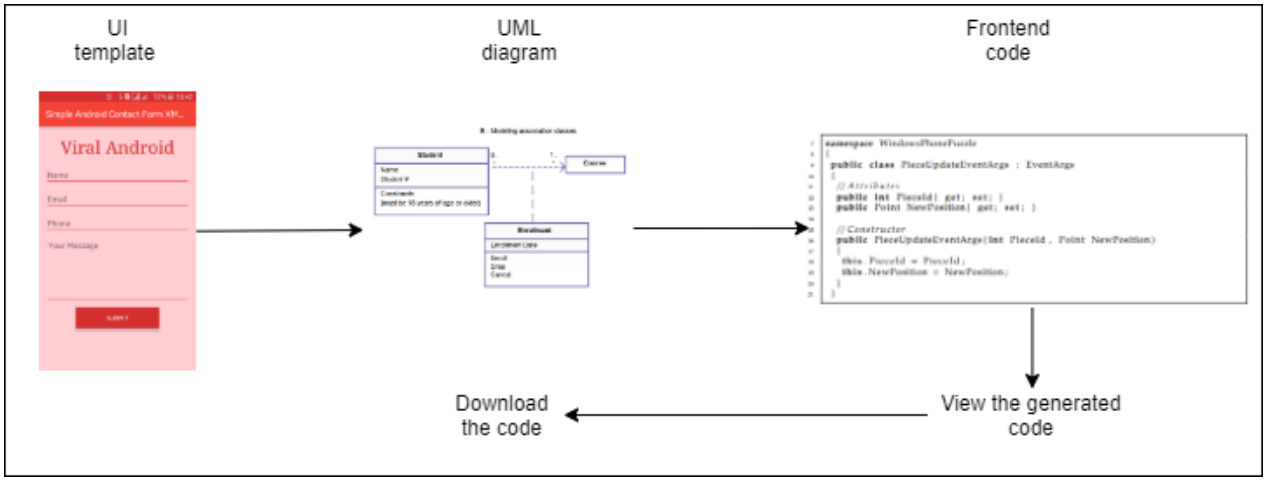
MEMBER 4	**D.P. Kavindi Gimshani - IT19150826**

Component – Mobile Frontend-code generator

The selected UI template from the component 1 will be implemented as the frontend of mobile application by this component.

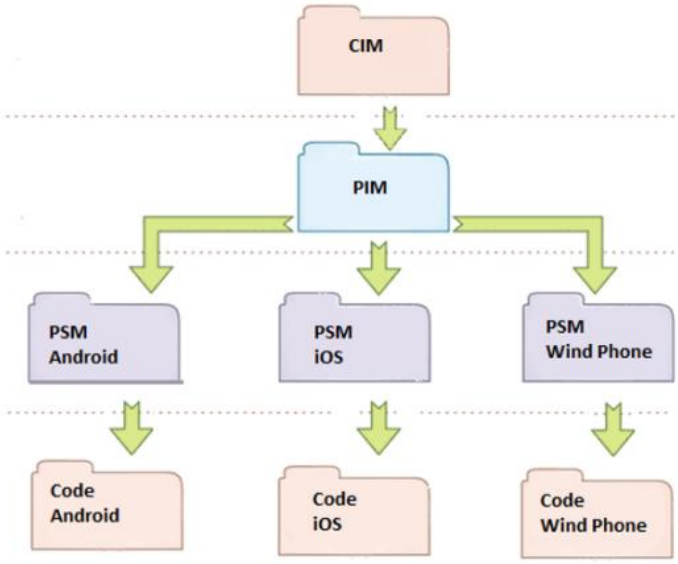
The process of the component -

1. Identify the components of the selected UI.
2. Generate the UML class diagram for the identifies components of the selected UI.
3. Generate the code for the frontend of the mobile application by using **Model Driven Architecture**.
4. Let the user to view the UI of generated code.
5. Download the generated code.



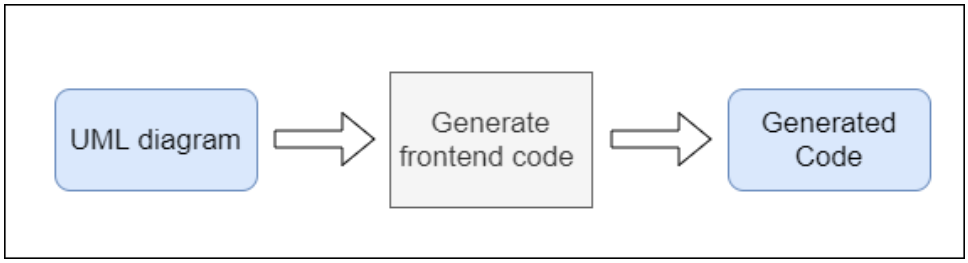
Related topics -

MDA defines three levels of models, Computation Independent Models (CIM), Platform Independent Models (PIM), and Platform Specific Models (PSM) for modeling the application and the successive transformations to automatically generate source code.



Explanation on input output for the component -

The input of the process is the UML class diagram. It goes through the process of MDA approach to generate the code based on UML class diagrams and gives the generated frontend code as the output.



Note -
The UML class diagram will be generated by the components and the requirements that are identified by the selected UI from component 1.

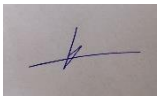
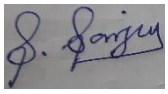
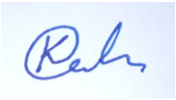
Advantages of this component -

- Code efficiency
- Interaction with device resources
- Responsive to multi devices

DECLARATION (Students should add the Digital Signature)

"We declare that the project would involve material prepared by the Group members and that it would not fully or partially incorporate any material prepared by other persons for a fee or free of charge or that it would include material previously submitted by a candidate for a Degree or Diploma in any other University or Institute of Higher Learning and that, to the best of our knowledge and belief, it would not incorporate any material previously published or written by another person in relation to another project except with prior written approval from the supervisor and/or the coordinator of such project and that such unauthorized reproductions will construe offences punishable under the SLIIT Regulations.

We are aware, that if we are found guilty for the above mentioned offences or any project related plagiarism, the SLIIT has right to suspend the project at any time and or to suspend us from the examination and or from the Institution for minimum period of one year".

	STUDENT NAME	STUDENT NO.	SIGNATURE
1	Ekanayake K.L.W (GROUP LEADER)	IT19150758	
2	Sakthivel.S	IT19158228	
3	H.M. Kasuni Navodya	IT19144986	
4	D.P Kavindi Gimshani	IT19150826	